What Drives Search Interest Around F1 Grand Prix Weekends?

1. Project goal and summary

I find that search interest in Formula 1 spikes around race day regardless of how close the title fight is, and that the level of attention stepped up after 2021.

The question. Every few weeks Formula 1 arrives in a new city and, for a few days, a large audience looks it up. I asked what drives that spike, and in particular whether a tight championship fight makes it bigger. The intuition is natural: suspense should draw attention. If it were true, promoters, broadcasters and host cities could expect more from races late in a close season. I also asked two questions about outside shocks. Did *Drive to Survive* (from 2019) raise the general level of interest, and did interest hold up in the 2020 pandemic season?

The project began somewhere else. I first planned to measure local economic impact through hotel and Airbnb prices in host cities. Hotel sites block scraping and Inside Airbnb covers only 36 of my races, so I moved to Google Trends search interest, which exists for every race and measures attention directly.

What I did. I built a pipeline that collects daily Google Trends series for 213 race weekends from 2016 to 2025 (including 2020), combines them with the points gap between the top two drivers before each race, and analyses them with hypothesis tests, frequent pattern mining, three clustering algorithms, two dimensionality reduction methods and five outlier detectors.

What I found.

- **A measurement flaw almost gave me a false answer.** My first spike metric, peak divided by baseline, correlated with closeness ($\rho = -0.18$, $p = 0.009$). It mostly measures how small the baseline was, and close seasons were early seasons with tiny baselines. The corrected metric, lift in index points, shows no relationship ($\rho = +0.04$, $p = 0.61$; pre-specified within-season test $r = -0.05$, $p = 0.52$).
- **Closeness is not the driver.** Every cut of the data I tried agrees, and the intervals rule out effects larger than about one index point out of a typical 21.
- **Attention is a three-day event.** Race day is the peak in 210 of 213 races, and the day before, race day and the day after carry 77% of the volume in a fifteen-day window.
- **There was a step in 2022, not 2019.** Median lift rose by 2.5 points (95% interval 1.6 to 3.7) and baseline interest roughly quadrupled from 2022. That is compatible with a *Drive to Survive* era effect but does not isolate it. The 2020 season shows no clear change.
- **Race weekends form three soft types** (silhouette 0.37) that follow circuit type and season, not the title fight: established year-round interest (street circuits such as Monaco), high-surge, and event-only.
- **Outliers are mostly artifacts.** Of 11 consensus outliers, three Las Vegas races are explained by Saturday-night timing, which I verified by shifting the window; two São Paulo races look like very low search volume.

Why it matters. The result is useful as a caution as much as a finding: era effects can pass as drivers, and a ratio on a self-scaled index can invent effects. For practitioners, benchmark each race against its own era and expect the attention window to last about three days. Sections 7 and 8 set out what these data cannot show, mainly total audience size, which needs a fixed reference scale across races.

2. Data

The dataset covers 213 Grand Prix weekends from 2016 to 2025, built from three public sources and one small supplementary source.

Source	What I took	Coverage
Wikipedia season pages	Race calendar: round, Grand Prix, circuit, date	214 races, 2016–2025 including 2020
Google Trends (pytrends, worldwide)	Daily search interest for the Grand Prix name, 60 days before to 30 days after each race	213 of 214 races
Jolpica-F1 API	Driver standings before each race, giving the points gap between first and second	203 races
Inside Airbnb (supplementary)	One listings snapshot per city: mean price and listing count	36 races, 2022–2025 only

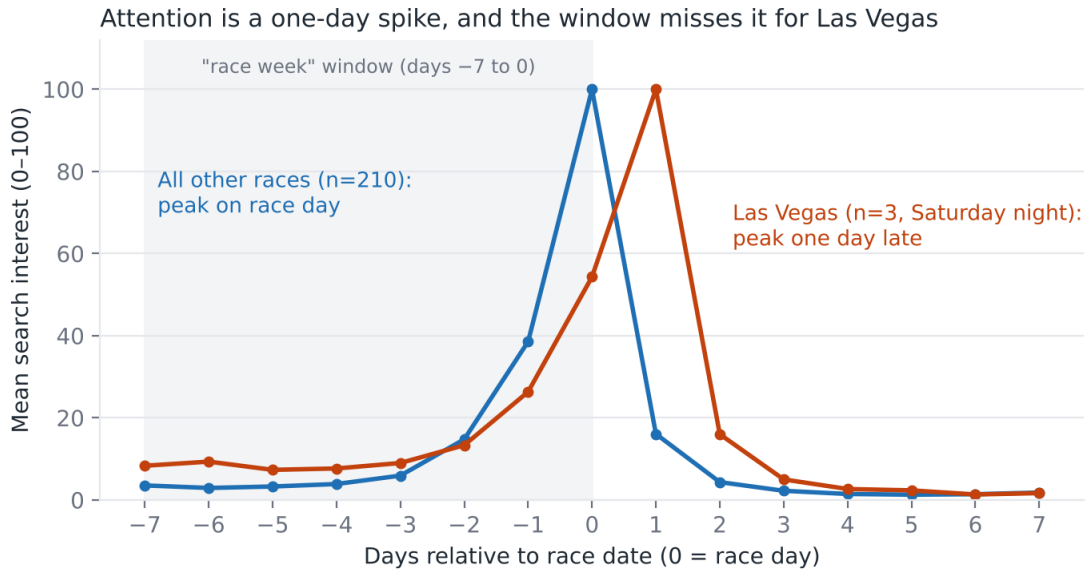
Why this window. Starting in 2016 gives three seasons before Netflix's *Drive to Survive* (first released in March 2019). I keep 2020 and flag it with an indicator variable, because a pandemic season with closed grandstands is a natural test of whether interest depends on being there in person.

Integration and match loss. The raw scrape had 224 table rows; removing footnote and footer rows left 214 real races. Trends returned data for 213. The 2021 São Paulo Grand Prix returned nothing after repeated retries and is the only race lost. The championship gap exists for 203 races: the ten season openers have no prior standings, so that gap is missing by construction rather than by error. Airbnb data exist for only 36 races. The Airbnb file covers the 92 races of 2022–2025; 43 of them matched a city that Inside Airbnb covers, and 36 had a usable listings file. Small host towns (Spielberg, Zandvoort, Imola) and several others have no coverage at all.

Season	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Races analysed	21	20	21	21	17	21	22	22	24	24

What each race looks like. For every race I use the 91-day series. The baseline is days −60 to −14 (47 days), and race week is days −7 to 0. I also keep the days before and after for shape features used later. Figure 1 shows the average series: interest sits near the floor, climbs over the two days before the race, and peaks on race day. In 210 of 213 races the single highest day is race day itself. The three exceptions

are all Las Vegas races, which are run on Saturday night local time and peak the next day (section 6 returns to this).



Descriptive statistics.

Variable	Mean	SD	Median	Range
Baseline interest (index points)	0.41	0.47	0.19	0.00–2.62
Race-week lift (index points)	21.14	3.19	20.66	11.20–32.92
Race-week share of 91-day volume	0.74	0.10	0.76	0.31–1.00
Old spike ratio (peak ÷ baseline)	739	1,148	214	0–4,700
Points gap between 1st and 2nd (n = 203)	41.99	48.92	24	0–276

Two features of these numbers shape everything that follows. First, baselines are tiny: the typical race has interest of 0.2 on a 0–100 scale outside the event, so "attention" here is almost entirely an event phenomenon. Second, lift barely varies (SD 3.2 around a mean of 21), because the series is scaled to its own peak and the peak falls on race day. Lift therefore mostly measures how much of the neighbouring days rise with the peak (how sharp the spike is), not how large it is in absolute searches. I return to this limit in sections 7 and 8.

The gap is skewed. Half of the 203 races had a gap under 24 points, and 103 are "close" under my 25-point definition versus 100 not close. The Airbnb listings are a single scrape, so the 36 price values

(mean $225 per night, range $146–$392) describe cities, not race weekends, and use approximate exchange rates. I treat them as a minor supplementary variable.

3. Measuring a spike: why the first metric failed

My first spike metric, race-week peak divided by baseline, mostly measured how small the baseline was, so I replaced it with a bounded lift in index points.

Every Google Trends pull is rescaled so its own busiest day equals 100. The peak in a race window is therefore almost always 100, and the ratio collapses to roughly 100 divided by the baseline. A race whose baseline averaged 0.02 index points received a "spike" of 4,700, while a race with a baseline of 20 received 5. The old metric ranked races by how obscure their search term was before the event, which says little about the event itself. In the data this showed up as a clear negative rank correlation between the old spike factor and the baseline ($\rho = -0.44$, Figure A1), against $\rho = +0.13$ for lift, and as a handful of identical 4,700 values at the top of the ranking.

I replaced it with three bounded measures, all computed from the daily series of each race:

- **Lift (index points)** = mean interest in race week (days −7 to 0) minus mean interest in the baseline (days −60 to −14). This is the main outcome.
- **Lift z-score** = lift divided by the baseline standard deviation, floored at 1 so a flat baseline cannot inflate it.
- **Race-week share** = race-week volume divided by the volume of the whole 91-day window, a 0 to 1 measure of how concentrated attention is.

Lift answers a cleaner question: how many index points does attention rise when the race arrives? Because Trends is scaled per race, lift is a within-race quantity. I compare lifts across races, but I do not read them as absolute search counts. There is a further consequence: since the race-day value is almost always 100, it contributes the same 12.5 points to every race's lift, and the differences between races come from how elevated the neighbouring days are. Lift is therefore best read as how sharply attention is concentrated, and section 7 states what that does and does not let me conclude.

Why this matters for the research question. With the old ratio, any variable that correlates with the baseline (era, host country, circuit age) appears to "explain" spikes. Early-era races with tiny baselines looked like the most dramatic events in the dataset, yet on lift they sit slightly below the later seasons. Fixing the outcome variable was the most consequential methodological decision in the project, because it changed which conclusions were available.

4. Hypothesis 1: does a tight title fight raise the spike?

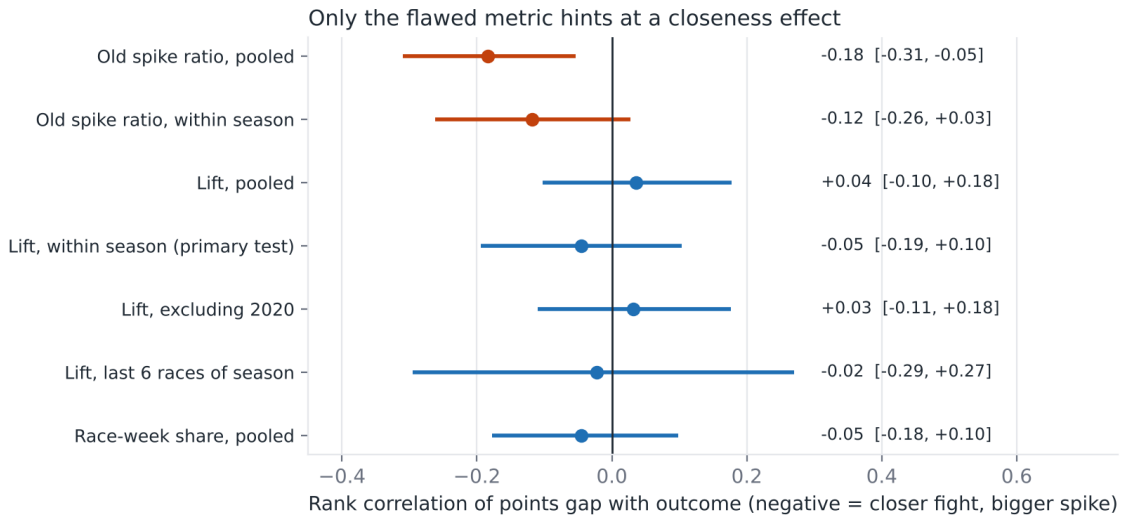
No: across every test I ran, championship closeness has no detectable relationship with race-week lift, and the confidence intervals rule out anything but a small effect.

Setup. Closeness is the points gap between first and second in the drivers' championship entering the race (smaller means closer). "Close" means a gap under 25 points, roughly one win. Before adding the 2016–2021 seasons I fixed one primary test: the rank correlation between gap and lift *within* each season, which removes differences between years. Everything else is exploratory, so I read p-values near 0.05 as suggestive only.

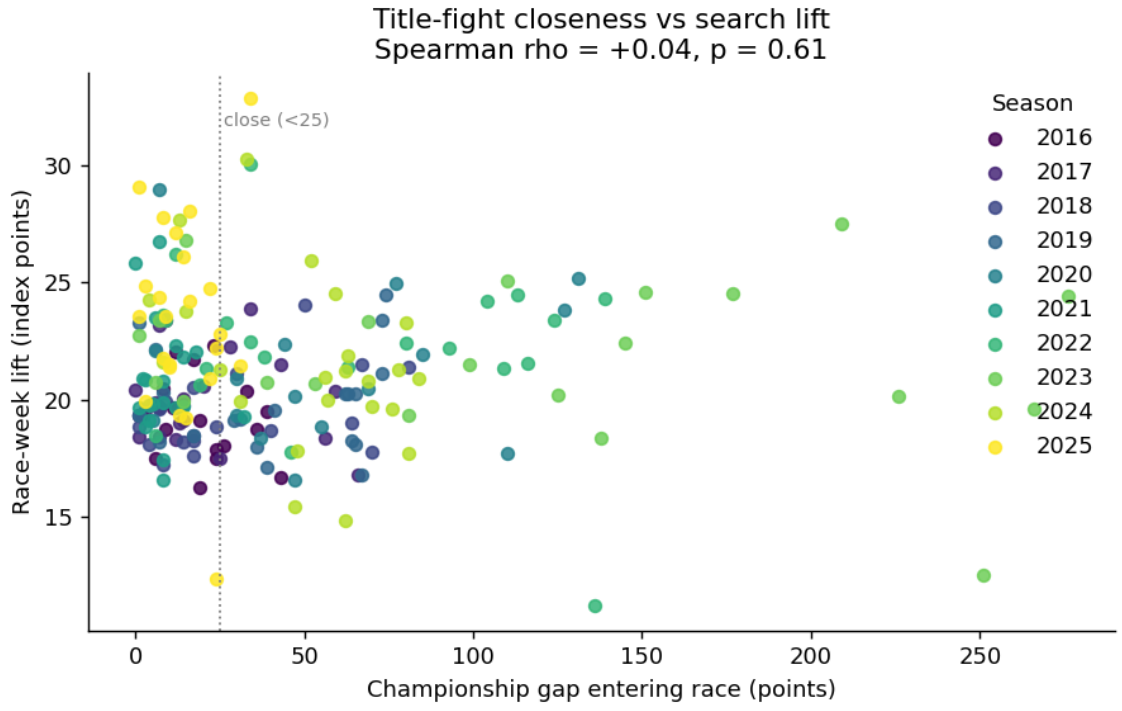
Results (n = 203 races with a gap).

Test	Result
Primary: within-season rank correlation, gap vs lift	r = −0.05, p = 0.52, 95% CI [−0.19, +0.10]
Pooled Spearman, gap vs lift	ρ = +0.04, p = 0.61, 95% CI [−0.11, +0.18]
Pooled Spearman, gap vs lift z-score	ρ = +0.06, p = 0.37
Pooled Spearman, gap vs race-week share	ρ = −0.05, p = 0.52
Season fixed-effects slope	−0.07 lift points per 10 gap points, 95% CI [−0.21, +0.06], permutation p = 0.24
Close (<25) vs not close, median lift	20.31 vs 20.84, Mann-Whitney p = 0.94
Same split at gap < 10 and gap < 50	p = 0.90 and p = 0.17
Excluding 2020	ρ = +0.03 (p = 0.67); primary r = −0.06 (p = 0.41)
Last six races of each season only (n = 59)	ρ = −0.02, p = 0.86
Title mathematically out of reach (about 21 races), median lift	21.5 vs 20.5 for open title fights, p = 0.38
Tight seasons (2017, 2021, 2025) vs blowout seasons (2020, 2023, 2024)	20.83 vs 21.21, p = 0.55
Is a lift anomaly more likely in a close race? (Fisher exact)	odds ratio 0.86, p = 0.81

Per season, the correlation runs from −0.38 (2024, p = 0.07) to +0.31 (2018) with five negative, four positive and one zero. No season is significant, and there is no consistent direction. The decided-title row uses an approximation (26 points available per remaining race, ignoring sprint races), so I report it as a robustness check, not a main result.



The old metric would have confirmed the hypothesis. With the original spike ratio, the pooled gap correlation is ρ = −0.18 (p = 0.009), apparently showing that close fights produce bigger spikes, and my frequent-pattern analysis in section 6 finds the same pattern. Two things explain it. The close-title seasons are concentrated in 2016–2021 (59% of those races were close, against 40% in 2022–2025), and those seasons had the lowest baselines (median 0.09 vs 0.46). A ratio of peak to baseline is largest where the baseline is smallest, so era alone produces a link between closeness and spike size. Within seasons the old-metric correlation falls to −0.12 and its interval includes zero. This is a concrete example of a confounded measure producing a plausible but false finding.



How large an effect could be hiding? With 203 races the study has about 80% power to detect a rank correlation of roughly 0.2. The season fixed-effects interval translates this into search terms: moving from an average gap to one standard deviation (49 points) tighter would change lift by 0.35 points, and by at most about 1 point at the interval's edge. Typical lift is 21 points, so a real closeness effect, if one exists, is under 5% of the race-week signal.

What this means in practice. A broadcaster, promoter or sponsor should not expect a close points battle to raise how sharply the public searches for a race weekend. The weekend itself carries the attention. There is one important qualification. Lift is a shape measure (section 7), so this test shows closeness does not change how concentrated and sharp the spike is. It cannot show whether a title fight increases the total audience, which would need a fixed reference scale across races.

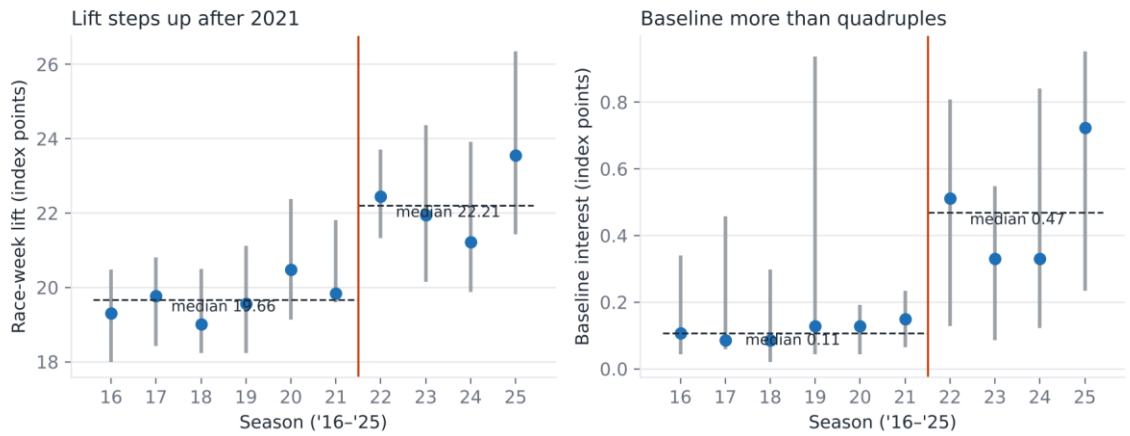
5. Hypothesis 2: Drive to Survive, the 2022 step, and 2020

Race-week lift and baseline interest both stepped up, but the step falls between 2021 and 2022, not when *Drive to Survive* launched in 2019, and the 2020 pandemic season shows no change.

The pre-specified comparison. Splitting at the show's launch, seasons up to 2018 against 2019 onward, gives significant differences: median baseline 0.09 vs 0.26 ($p = 0.015$), median lift 19.41 vs 21.32 ($p < 0.001$), and race-week share 0.80 vs 0.75 ($p = 0.006$). Taken alone this reads like a Netflix effect. A scan over every possible break year shows the story is more specific.

First season of the "after" period	Median lift before → after	p (lift)	p (baseline)
2019 (show launch)	19.41 → 21.32	3e-6	0.015
2020	19.49 → 21.53	4e-8	0.011
2021	19.57 → 21.62	1e-7	4e-4
2022	**19.66 → 22.21**	**6e-9**	**9e-6**
2023	19.94 → 21.82	7e-6	2e-4

The break at 2022 fits best on both lift and baseline. Median lift is 2.5 points higher in 2022–2025 than in 2016–2021 (bootstrap 95% interval 1.6 to 3.7). Median baseline is 0.09 up to 2018, 0.13 in 2019–2021 and 0.47 from 2022 on, so most of the rise in off-event interest comes in 2022, not 2019. The seasons within each period do not differ from each other (Kruskal-Wallis $p = 0.29$ within 2016–2021 and $p = 0.18$ within 2022–2025). It looks like a step, not a gradual trend. Lift in 2019–2021 is only weakly higher than before 2019 ($p = 0.06$).



Dots = season median, bars = interquartile range. 2020 shown as an ordinary season. Orange line marks the break between 2021 and 2022.

Season	Races	Median lift	Median baseline	Median gap
2016	21	19.31	0.11	18
2017	20	19.78	0.09	14
2018	21	19.02	0.09	17
2019	21	19.56	0.13	40
2020	17	20.48	0.13	47
2021	21	19.83	0.15	7.5
2022	22	22.45	0.51	46
2023	22	21.94	0.33	110
2024	24	21.23	0.33	57
2025	24	23.56	0.72	13

What the step does and does not tell me. The data are consistent with the show's reach building over its later seasons, and 2022 also brought a new US race (Miami) with Las Vegas following in 2023. But the same years differ in many other ways, including new technical regulations and wider social-media reach, and I have no variable that separates them. I therefore describe a 2022 step in attention that is compatible with a Drive to Survive era effect, and I do not claim the show caused it. One further caution is measurement: Trends values for low-volume terms are coarse, so part of the step could reflect how Google reports small numbers in earlier years.

Race-week share falls after 2018 while baseline rises. In other words attention became less purely a race-week phenomenon: more of the 91-day volume is spread across the weeks around the event. That is what a growing year-round audience would produce.

The 2020 season. The 17 races of 2020 show a median lift of 20.48 against 20.69 for all other seasons ($p = 0.94$), and 2020's season median is close to those of 2019 (19.56) and 2021 (19.83), so there is no break there. Baseline was lower (0.13 vs 0.23, $p = 0.050$) and race-week share higher (0.82 vs 0.76, $p =$

0.053), both borderline. With 17 races, only large changes are detectable, so the right reading is that the pandemic season left no clear mark on search interest, not that it had no effect.

What this means in practice. For anyone planning around F1 attention (sponsors, host cities, broadcasters), the large movement in the data is a one-time rise in the level of interest, not a race-by-race effect of competitive drama. Any model that pools seasons without accounting for this will credit the era's shift to whatever variable happens to differ between early and late seasons, as the old metric did with title closeness. The practical rule is to benchmark each race against its own season or era.

6. Pattern mining, clustering, dimensionality reduction and outliers

Race weekends fall into three soft attention types that follow era and circuit type, not the title fight, and the apparent outliers turn out mostly to be measurement artifacts I can partly verify.

6.1 Frequent pattern mining

I turned each race into 23 true/false columns: median splits for baseline and spike size, "close championship" (gap under 25), "has Airbnb data", season and circuit type (street or permanent) one-hot encoded, and the 2020 flag. I ran Apriori and FP-Growth at three minimum supports.

Minimum support	Itemsets (both algorithms)	Apriori runtime	FP-Growth runtime
0.05	106	4.6 ms	30.1 ms
0.10	21	2.4 ms	5.0 ms
0.20	9	3.0 ms	3.8 ms

Both algorithms return identical itemsets, as they must, and Apriori is faster at every level. FP-Growth's advantage appears on much larger tables; with 213 rows and 23 columns the difference is milliseconds and does not matter. At support 0.10 the rules are weak: the highest lift is 1.53. Three are worth reading.

- **Street circuit ↔ high-baseline city** (lift 1.53). 76% of street races fall in the higher-baseline half. Street races (Monaco, Singapore, Baku, Las Vegas) are staged in or near cities that people search for all year, so their search term carries interest outside the event.
- **Permanent circuit + close championship ⇒ spike above median** (support 0.26, confidence 0.70, lift 1.41). This is the rule that appears to support Hypothesis 1. It uses the old peak-over-baseline ratio, and it is the same era artifact described in section 4: close seasons are early seasons with tiny baselines.
- **Street circuit ⇒ close championship** (confidence 0.63, lift 1.31). I have no mechanism for it and treat it as incidental.

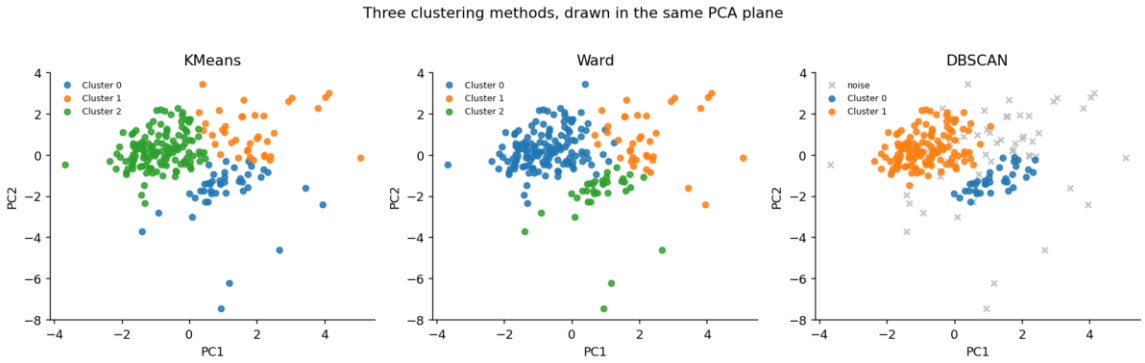
Pattern mining is useful here mainly as a warning: it found a plausible rule that the later tests show to be an artifact of the outcome variable.

6.2 Clustering

Features. Each race is described by the shape of its search curve: log baseline, lift, race-week share, build-up (mean of days −13 to −8 minus baseline) and post-race drop (race-week mean minus the mean of days +1 to +7). I standardized all five, because distance methods otherwise follow whichever feature has the largest units. Context variables (championship gap, season, circuit type) are not used to form clusters; I use them afterwards to ask what the clusters mean.

Choosing the number of clusters. For K-means, silhouette scores for k = 2 to 8 are 0.356, 0.373, 0.356, 0.306, 0.310, 0.281 and 0.249. I pick K = 3, the best among k = 3–6 (k = 2 is the coarsest split and nearly as good). Ward hierarchical clustering with K = 3 gives a silhouette of 0.366 and a cophenetic correlation of 0.58. DBSCAN, with minPts = 5 and ε = 0.74 chosen by a grid search, finds two dense clusters and labels 53 races (25%) as noise.

The methods agree where it matters. K-means and Ward are very similar (adjusted Rand index 0.82). DBSCAN agrees less (ARI 0.65 with K-means, 0.64 with Ward) because it declares a diffuse group to be noise rather than forcing it into a cluster: 29 of the 38 "high-surge" races are DBSCAN noise. A silhouette near 0.37 says the groups overlap; I treat them as descriptive types.



K-means cluster	Races	Typical race weekends	Profile	Share of 2022–25	Street circuits	Median gap
0: Established, year-round interest	43	Monaco 2018, 2019, 2023	Highest baseline; lowest race-week share (0.60); lift 19.7	56%	51%	30
1: High-surge	38	Australia 2024 and 2018, Azerbaijan 2023	Largest lift (25.6), largest build-up and steepest post-race	76%	11%	16

K-means cluster	Races	Typical race weekends	Profile	Share of 2022–25	Street circuits	Median gap
			drop			
2: Event-only	132	Azerbaijan 2018, Canada 2022, Bahrain 2021	Lowest baseline; race-week share 0.80; lift 20.3	30%	9%	24

What the clusters line up with. The clusters do not differ by championship gap (Kruskal-Wallis p = 0.68) or by the share of close races (chi-square p = 0.55). They differ strongly by circuit type (street share, p < 0.001) and by season (p = 0.005), and weakly by 2020 (p = 0.07). So the main structure in these attention curves is who the host is and when the race was run. The title fight plays no visible part, which matches the answer to Hypothesis 1 from a completely different method.

Practical reading. A race promoter can place a new weekend into one of these types and compare it with its type, not with the whole calendar. A street-circuit race with a baseline well under its type's norm is underperforming for its kind; an event-only race with a large build-up is doing something unusual. I offer this as a benchmarking idea. The clusters are soft, so any real use would need a fixed reference scale (section 8).

6.3 Dimensionality reduction

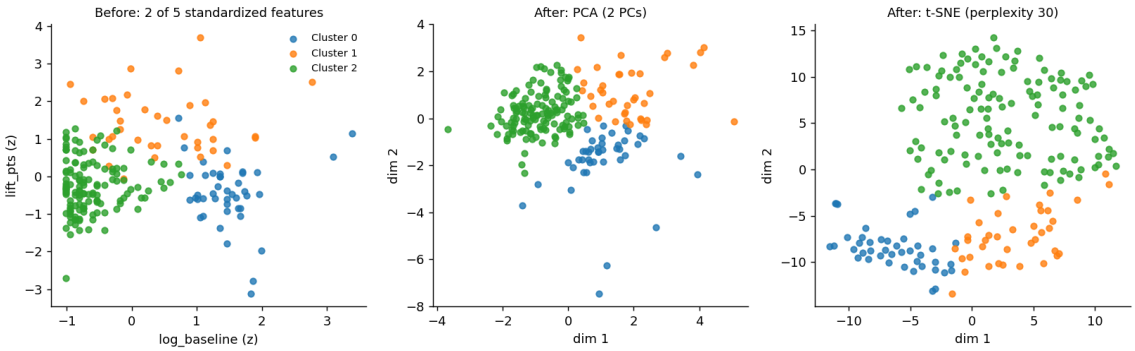
The five shape features are highly redundant, so two principal components keep 81% of the information and clustering on them barely changes.

Redundancy. Log baseline and race-week share are strongly negatively correlated (r = −0.85): a higher baseline leaves a smaller share of volume to the race week. Lift and post-race drop are strongly positively correlated (r = 0.81). Compression is therefore natural.

PCA. The first two components explain 41.0% and 39.9% of the variance (80.9% together), the third adds 14.4% (95.3% cumulative), and the last two add under 5%. The first component is the overall *level of attention outside the peak*: it loads positively on baseline (0.52), build-up (0.49) and lift (0.45) and negatively on race-week share (−0.45). The second is *how sharp the peak is*: it loads on drop (0.59), race-week share (0.52) and lift (0.50).

t-SNE. I added a non-linear method as a check. With perplexity 30 it keeps local neighbourhoods better than PCA (trustworthiness 0.975 vs 0.941) but distorts global distances (Spearman correlation of pairwise distances 0.76 vs 0.96 for PCA). I use it only to look at the data, because distances between t-SNE groups carry no meaning.

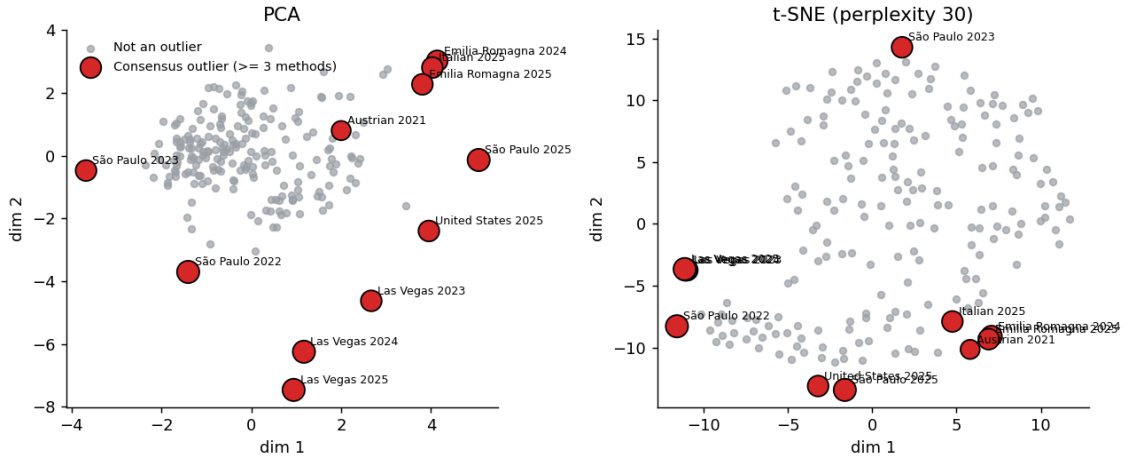
Effect on clustering. K-means on the first three components reproduces the full five-dimensional clusters almost exactly (ARI 0.98), and so does K-means on two components (ARI 0.98). K-means on the t-SNE coordinates gives quite different groups (ARI 0.38), which confirms t-SNE is a picture, not a space to cluster in.



6.4 Outlier detection

Definition. An outlier is a race weekend whose five-feature profile is unusual compared with the others: far from its neighbours, or extreme on one feature. An unusual profile can be a rare real event, a measurement artifact of Google Trends, or a genuinely different kind of weekend. Telling these apart is the point of the analysis.

Methods. I used five detectors, each set to flag about 5% of races: a z-score rule (any feature beyond 3 standard deviations), Isolation Forest, Local Outlier Factor, mean distance to the five nearest neighbours, and One-Class SVM. They flagged 12, 11, 11, 11 and 15 races. Agreement between them is high among the neighbour-based and tree-based methods (overlap 0.64–0.83) and low for One-Class SVM (0.23–0.37), so I require a race to be flagged by at least three of the five. That gives 11 consensus outliers (5.2% of races). Every one of them is also noise in DBSCAN, so a density-based method independently agrees.



Race	Votes	What is unusual	What I found
Las Vegas 2023, 2024, 2025	4–5	Very negative post-race drop	**Verified artifact.** Races run on Saturday night, so the search peak falls on day +1, outside my window. Moving the window by one day

Race	Votes	What is unusual	What I found
			raises lift from 19.6, 14.8 and 12.3 to 30.7, 26.5 and 23.9.
São Paulo 2022, 2023	4–5	Among the lowest lifts in the data (11.2, 12.5)	Interest is zero on the days either side of race day (a one-day spike over a nearly empty series), which points to low search volume rounded to zero. The cause (term dilution or low volume) cannot be separated with these data.
São Paulo 2025	5	Low race-week share	Likely genuine: a high baseline (2.0) and a lift of 29 in a race entered with a one-point title gap.
Emilia Romagna 2024, 2025	4	Very strong build-up	Not explained. Both have small baselines (0.3), so a few days of elevated interest in the run-up look large.
Italy 2025	4	Highest lift in the data (32.9)	No artifact found; a rare, strong weekend.
United States 2025	4	Highest baseline (2.62)	The term carries year-round interest; a term effect, not a race effect.
Austria 2021	3	Strong build-up	Not explained.

Only five of the eleven occurred in a close-title race, against 51% of all races, which is not an unusual share. The most valuable result of this section is the Las Vegas row. I had guessed the Saturday-night explanation, then tested it: the three Las Vegas races are the only ones in the data whose peak is not on race day, and shifting the window removes the anomaly. Those three lifts were understated, and they should be recomputed in any future version.

Impact of outliers on the other methods. Removing the 11 races lowers the K-means silhouette from 0.373 to 0.341 on the remaining 202 and re-assigns a share of races (ARI 0.53 between the old and new

clusterings), because the "high-surge" cluster is a loose tail and loses members. PCA is unaffected (80.9% of variance with outliers, 81.4% without). The link between cluster and championship gap stays non-significant without them ($p = 0.09$, from 0.68). Outliers therefore soften cluster boundaries but do not alter the structure or any conclusion.

7. Discussion: what drives the spike, and what I can learn

The spike is driven by the race itself and by the era it happens in, and almost not at all by how close the title fight is.

What explains the variation I can measure. Race day is the peak in 210 of 213 races, and on its own it contributes 12.5 of the 21.1 index points of the average lift (100 points on one day out of eight). The rest of the variation is small and structured. A single indicator for the years from 2022 explains about 16% of the variance in lift ranks, and a full set of season indicators about 20%. Adding the championship gap on top of seasons raises explained variance from 22.1% to 22.4%, and the gap alone explains 0.1%. A joint rank regression of lift on baseline, gap, round number, street circuit and the 2020 flag explains 1% in total. In short, when the race happened matters a great deal more than how tight the points were.

Why I believe the null on closeness. Three independent lines point the same way. Every test of the gap against lift, share and lift z-score is flat, including the pre-specified within-season test and its season fixed-effects version. Clusters built without the gap do not differ by gap. And the one analysis that appeared to find an effect, the old spike ratio and the matching association rule, depends on an outcome that mostly measures a small baseline, and it weakens to nothing within seasons. The confidence intervals also exclude effects larger than about one index point.

Where closeness could still matter. My instrument has two blind spots. First, lift is a shape measure: because every pull is scaled to its own peak, it tells me how sharply attention concentrates around race day, not how many people search. A title fight could enlarge the audience without changing the shape, and I could not see that. Second, I search for the Grand Prix name. Interest in a title fight might show up in searches for the drivers or for the championship, which I did not collect. The honest claim is narrow: closeness does not change the sharpness of the spike in searches for the race.

Findings worth carrying forward.

1. **Attention is brief.** Averaged over all races, the three days from the day before to the day after the race hold 77% of the volume in the surrounding fifteen-day window. Race day reaches 99, the day before 38, the day after 17, and two days later 4.5. Any activity that depends on search-driven attention (promotion, content, ticket sales, local sponsor activation) has a window of about three days.
2. **There was a step in 2022.** Baseline interest roughly quadrupled and lift rose by about 2.5 points. Benchmarks drawn from before 2022 understate present-day attention, and models that mix eras need an era term.
3. **Types are about place and era.** The three attention types separate street circuits from permanent ones and early seasons from late ones. A host city is best compared with its own type.

4. **Measurement choices can create findings.** The most important result for method is that the first metric supported my hypothesis and the corrected one did not. The corrected one is the one to believe, because I can explain why the first was wrong.

How confident am I? Most confident: the null on closeness, the race-day peak and the 2022 step. Moderately confident: the three clusters, which are soft and overlap. Least confident: the explanations for individual outliers, except Las Vegas, which I verified. The Airbnb layer is too thin to support any claim and I do not draw one.

8. Limitations and next steps

The main limitation is the instrument: Google Trends gives a relative, per-race index, so I can compare shapes and lifts but never absolute audiences.

Measurement limits

- **Per-race scaling.** Each pull is scaled to its own maximum. A lift of 20 points means a large rise relative to that race's own peak, not 20 units of searches. A fixed reference term in every pull would let me anchor all races on one scale.
- **Search term ambiguity.** The query is the Grand Prix name. Terms such as "Las Vegas Grand Prix" or "São Paulo Grand Prix" can be diluted by non-F1 searches, and an English query can miss local-language searches. I treat the São Paulo outliers as a hypothesis about this, not a result. The Las Vegas outliers I tested directly (see the Date alignment note below).
- **Date alignment.** Race week is days −7 to 0. A Saturday-night race (Las Vegas) can push the search peak into the following day, outside the window. I tested this for the three Las Vegas races by shifting the window, and the anomaly disappeared. I have not re-run the shifted window for every Saturday-night race.
- **Resolution at tiny volumes.** Trends rounds very small volumes to zero, so some baselines are effectively empty.
- **One missing race.** The 2021 São Paulo Grand Prix returned no data after repeated retries and is excluded.

Design limits

- **Era and treatment are confounded.** Seasons differ in rules, drivers, media rights, social media reach and Drive to Survive all at once. I can describe when attention stepped up, but not isolate why.
- **Closeness is measured by one definition.** I use the points gap between first and second before the race, with 25 points as the "close" cutoff, and check 10 and 50 as well. Other definitions (gap to the fifth driver, probability of the title changing hands) could behave differently.
- **Many tests.** The primary test was fixed in advance; everything else is exploratory, and p-values near 0.05 should be read as suggestive.
- **Soft clusters.** Silhouette scores near 0.4 describe overlapping groups, so cluster labels are descriptions, not categories.
- **Thin economic layer.** Inside Airbnb data cover only the 2022–2025 races in a few large cities, so they cannot support conclusions about local economic impact.

Concrete next steps

1. Recompute lift with a one-day-shifted window for all Saturday-night races, not just Las Vegas, and correct the three understated Las Vegas lifts.
2. Add a fixed reference term to each pull so lifts share one scale.
3. Use localized queries for non-English-speaking hosts.
4. Fit a model with season fixed effects, so closeness is tested only within seasons while era is absorbed.
5. Look at the daily series around specific events (a championship decider, a crash, a first win) to see what produces the largest day-level spikes.

9. Reproducibility

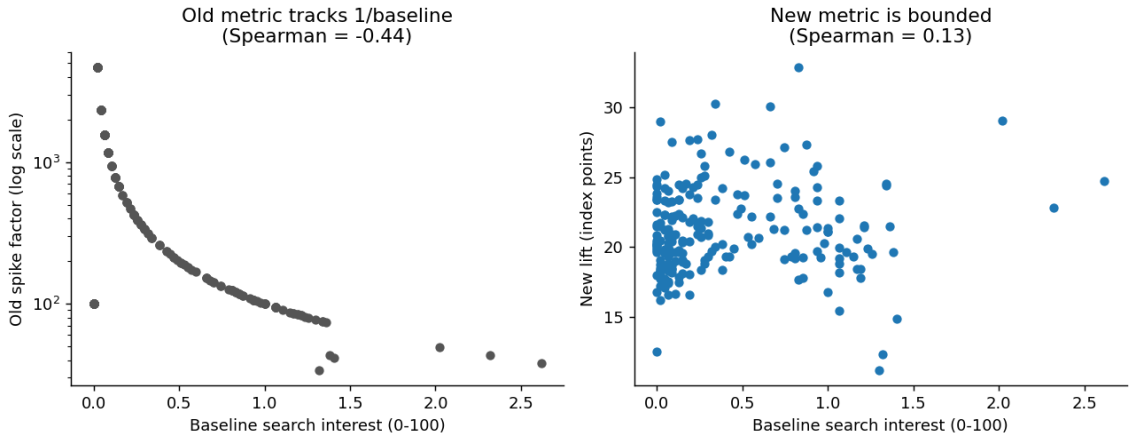
The whole analysis reruns from twelve numbered Python scripts, and cached downloads mean a rerun does not hit the web APIs again.

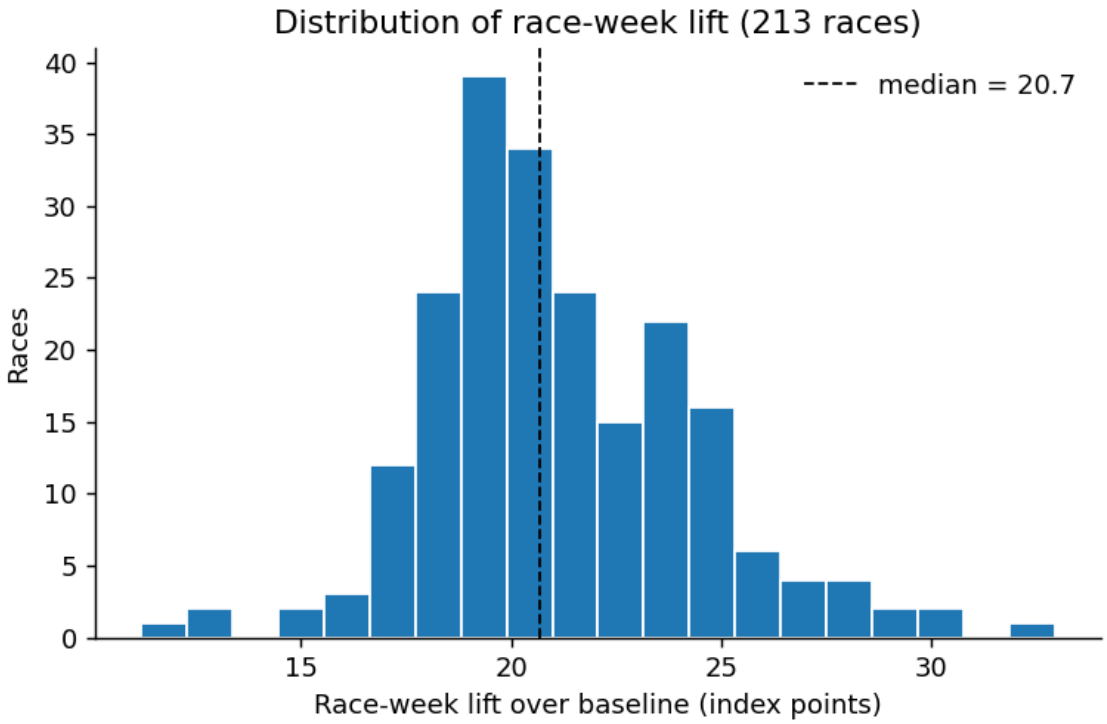
Script	What it does
01_scrape_races	Scrapes the Wikipedia season calendars for 2016–2025, normalizing column names across years
02_pull_trends	Pulls daily Google Trends per race (−60 to +30 days) with retries, backoff and a cache
03_preprocessing	Merges races and trends, builds the old spike metric, adds the championship gap from the Jolpica-F1 API (cached) and the COVID flag
04_anomaly_detection	First-pass anomaly flags (z-score, Isolation Forest, LOF) on the old metric
05_visualize_anomalies	Exploratory anomaly figures
07_missing_values_and_integration	Missing-value frequencies and match-loss counts at each integration stage
08_descriptive_stats_and_normalization	Descriptive statistics, exploratory figures, log transforms
09_binary_features	One-hot and threshold-based binary table for pattern mining
10_frequent_pattern_mining	Apriori and FP-Growth at several supports, association rules
11_final_analysis	New lift metric, hypothesis tests, driver analysis, figures A–G
12_clustering_dimred_outliers	K-means, Ward, DBSCAN, PCA, t-SNE, five outlier

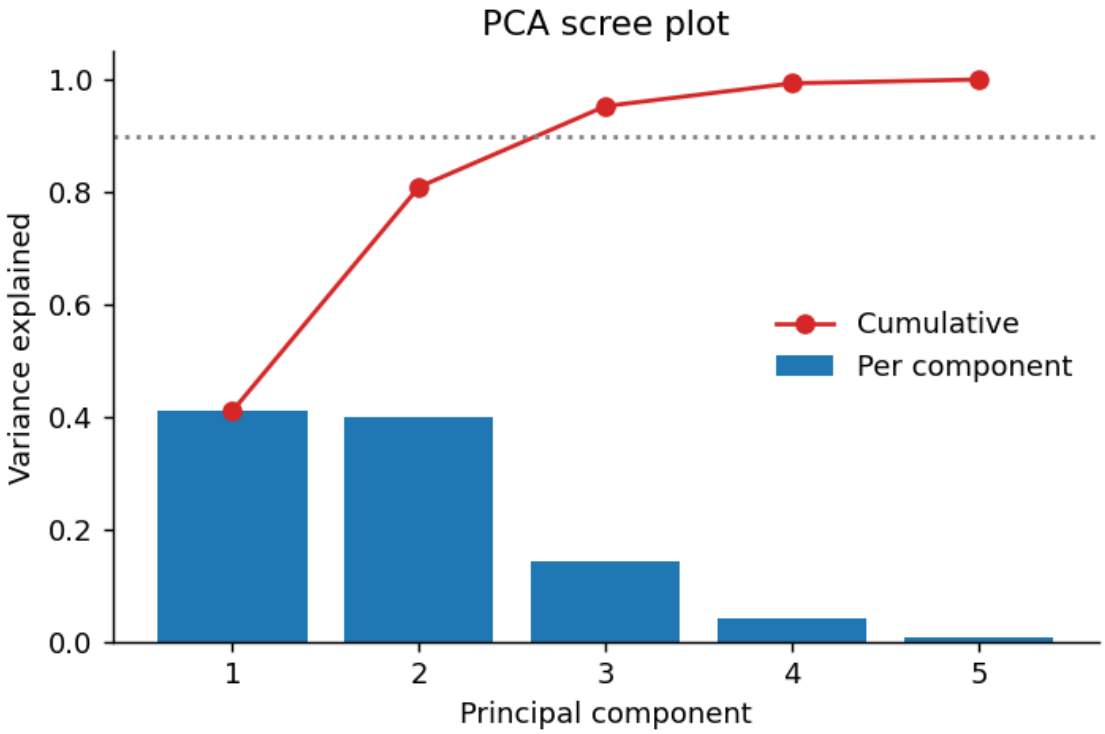
Script	What it does
	methods, impact of outlier removal

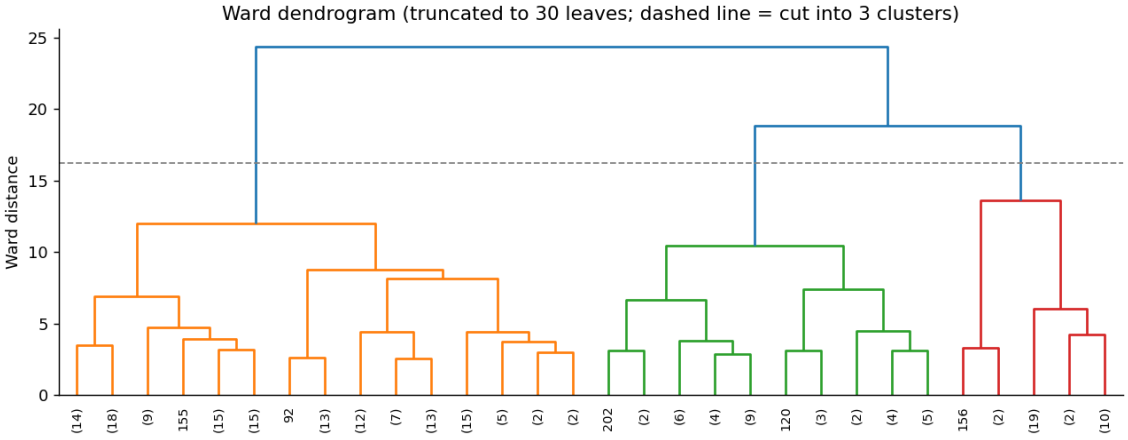
Script 06 (Airbnb) is run separately for the 2022–2025 races. All random procedures use seed 42.

Figure appendix






PCA scree plot
Variance explained
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
Principal component
Cumulative
Per component


Ward dendrogram (truncated to 30 leaves; dashed line = cut into 3 clusters)
Ward distance
25
20
15
10
5
0
(14)
(18)
(9)
155
(15)
(15)
92
(13)
(12)
(7)
(13)
(15)
(5)
(2)
(2)
202
(2)
(6)
(4)
(9)
120
(3)
(2)
(4)
(5)
156
(2)
(19)
(2)
(10)